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Nutrient demand interacts with legume maturity to affect rumen pool sizes in dairy cows

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ABSTRACT

Effects of legume maturity on dry matter intake (DMI), milk production, ruminal fermentation and pool sizes, and digestion and passage kinetics, and the relationship of these effects with preliminary DMI (pDMI) were evaluated using 16 ruminally and duodenally cannulated Holstein cows in a crossover design with a 14-d preliminary period and two 17-d treatment periods. During the preliminary period, the pDMI of individual cows ranged from 22.9 to 30.0 kg/d (mean = 25.9 kg/d) and the 3.5% fat-corrected milk yield ranged from 34.1 to 68.2 kg/d (mean = 43.7 kg/d). Experimental treatments were diets containing alfalfa silage harvested either a) early-cut, less mature (EC) or b) late-cut, more mature (LC) as the sole forage. Early- and late-cut alfalfa contained 40.8 and 53.1% neutral detergent fiber (NDF) and 23.7 and 18.1% crude protein, respectively. Forage:concentrate ratios were 53:47 and 42:58 for EC and LC, respectively; both diets contained approximately 22% forage NDF and 27% total NDF. Preliminary DMI, an index of nutrient demand, was determined during the last 4 d of the preliminary period when cows were fed a common diet and used as a covariate. Main effects of alfalfa maturity and their interaction with pDMI were tested by ANOVA. Alfalfa maturity and its interaction with pDMI did not affect milk yield but EC increased DMI compared with LC; thus, EC had lower efficiency of milk production than LC. The EC diet decreased milk fat concentration more per kilogram of pDMI increase than the LC diet, but milk fat yield was not affected. The lower concentration and faster passage rate of indigestible NDF for EC resulted in lower rumen pools of indigestible NDF, total NDF, and dry matter than did LC, which EC increased at a slower rate than did LC as pDMI increased. The EC diet decreased starch intake and increased ruminal pH compared with the LC diet. The rate of ruminal starch digestion was related to level of intake, but this

did not affect ruminal or postruminal starch digestion. Total-tract digestibility of NDF, organic matter, and dry matter was higher for EC than LC. Microbial efficiency tended to be related to pDMI and the response differed by treatment. When alfalfa silage was the only source of forage in the diet, cows supplemented with additional concentrate to account for decreased protein and increased fiber concentrations associated with LC produced similar fat-corrected milk yields with greater efficiency than cows fed EC.

Key words: alfalfa maturity, digestion kinetics, rumen pool

INTRODUCTION

The supply and utilization of nutrients in dairy cows fed forage-based diets are affected by forage maturity at harvest. Generally, postponing harvest increases DM yield, but nutritive quality decreases as the concentration of NDF increases and its digestibility decreases because of greater lignification. Dairy cows fed less mature forage generally produce more milk with less supplemental concentrate than cows fed more mature forage. Thus, harvesting and feeding quality forages may improve animal performance, decrease purchased feed costs, and increase profitability.

Fiber concentration and digestibility affect feed intake by contributing to physical fill in the rumen. As forages mature, the indigestible fraction of fiber increases and the rate of digestion of the potentially digestible fraction decreases (Smith et al., 1972). Additionally, highly lignified (i.e., more mature) forages increase retention time in the rumen due to a slow rate of digestion (Allen, 2000). This suggests that more mature forages result in slower ruminal turnover and greater ruminal distention. Therefore, the degree to which rumen fill limits feed intake increases when dairy cows are fed more mature forage compared with less mature forage. Alfalfa (**AL**; *Medicago sativa*) was selected as a representative legume for use in this experiment because it is the predominant legume fed to dairy cows in the United States.

In addition to dietary factors affecting ruminal passage, digestion, and distention, the individual cow's ap-

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Table 1. Characterization of 16 cows during the final 4 d of the 14-d preliminary period, when cows were fed a common diet

Parameter	Median	Mean	SD	Minimum	Maximum
Parity	3	2.88	0.79	2	4
BW, ¹ kg	557	569	43	528	692
BCS	2.22	2.24	0.38	1.63	3.06
DIM	155	137	45	56	208
Milk, kg/d	40.0	41.7	9.6	29.6	66.1
3.5% FCM, kg/d	42.4	43.7	8.6	34.1	68.2
DMI, kg/d	26.1	25.9	2.1	22.9	30.0

¹Empty BW (ruminal digesta removed).

petite will also affect the responses of passage rate and intake to forage maturity. Cows respond differently to treatments depending on their level of intake (Voelker Linton and Allen, 2008) and production (Oba and Allen, 1999). Because legume maturity and level of intake affect ruminal passage and digestion rates and, thus, the physical filling effects in the rumen, the response to effects of legume maturity and its relationship with intake level should be assessed to determine if responses to treatment vary among cows with a wide range in DMI. We hypothesized that responses of DMI and digesta passage rates to legume maturity are related to level of intake and earlier maturity will permit a greater increase in passage rate than later maturity as feed intake increases.

The objectives of this experiment were to evaluate the relationships between voluntary DMI and effects of legume maturity on DMI, milk production, ruminal fermentation and pool sizes, and digestion and passage kinetics in lactating dairy cows. This study had 3 distinctive features to improve our understanding of the role of legume maturity and interpret its effect on animal responses. First, it allowed effects of the interaction between legume maturity and preliminary DMI (**pDMI**) to be evaluated. The use of pDMI, an index of nutrient demand, allowed the evaluation of treatments on animal responses in relation to level of intake and provided an indicator to test effects of intake level independent of treatments. Second, it directly compared treatment effects of early- and late-cut AL as the sole source of forage. Third, ruminal passage rates of individual feed fractions, instead of entire feeds, were measured using ruminally and duodenally cannulated cows.

MATERIALS AND METHODS

Cows and Treatments

Experimental procedures were approved by the Institutional Animal Care and Use Committee at Michigan

State University (East Lansing). Sixteen multiparous Holstein cows from the Michigan State University Dairy Cattle Teaching and Research Center were assigned randomly to treatment sequence in a crossover design experiment with one 14-d preliminary period and two 17-d experimental periods. The first 10 d of each period were allowed for diet adaptation and samples were collected during the final 4 d of the preliminary period and 7 d of each experimental period. Cows were 137 ± 45 (mean $\pm$ SD) DIM at the end of the preliminary period and were selected to provide a wide range and uniform distribution of pDMI and milk yield. During the final 4 d of the 14-d preliminary period, the average pDMI among cows ranged from 22.9 to 30.0 kg/d (mean = 25.9 kg/d) and 3.5% FCM yield ranged from 34.1 to 68.2 kg/d (mean = 43.7 kg/d; Table 1). Prior to calving, cows were cannulated ruminally (Bar Diamond Inc., Parma, ID) and duodenally with a gutter-type T cannula placed approximately 10 cm distal to the pylorus (Joy et al., 1997). Surgery was performed at the Department of Large Animal Clinical Science, College of Veterinary Medicine, Michigan State University.

Experimental treatments were diets containing AL silage from the same field harvested either a) early cut, less mature (**EC**) or b) late cut, more mature (**LC**) as the sole forage. Alfalfa was produced at the campus farm at Michigan State University, harvested using a New Holland FP230 pull-type forage harvester (New Holland, Racine, WI) set according to manufacturer specifications for a theoretical length of cut of 10 mm, and ensiled in Ag-Bags (Ag-Bag Systems Inc., St. Nazianz, WI). During the sample collection periods, early- and late-cut AL contained 40.8 and 53.1% NDF and 23.7 and 18.1% CP, respectively (DM basis; Table 2). Diets EC and LC were formulated to contain 22% forage NDF, 30% total NDF, and 18% CP. We acknowledge that these treatments affect dietary starch concentration, but maintaining similar forage and total NDF concentrations for both treatments was of primary interest. The diet fed during the preliminary period was formulated so that early- and late-cut AL each

Table 2. Chemical composition, particle size distribution, and fermentation parameters of the early- or late-cut alfalfa silage included in the treatment diets

Item	Alfalfa silage	
	Early	Late
Chemical composition		
DM, %	30.2	35.6
OM, % DM	89.5	92.1
NDF, % DM	40.8	53.1
iNDF, ¹ % DM	27.7	39.3
iNDF, % of NDF	67.8	74.1
ADF, % of DM	37.4	44.8
ADL, % of DM	8.41	11.9
CP, % DM	23.7	18.1
Starch, % DM	1.16	1.48
NDF digestibility, ² %	28.8	24.4
Particle size distribution ³		
Wet sieving, % DM retained		
19.0 mm	8.61	8.84
9.50 mm	20.3	11.7
4.75 mm	33.4	33.9
2.36 mm	19.2	27.1
1.18 mm	7.32	7.52
0.600 mm	4.34	4.99
0.300 mm	2.81	2.57
0.150 mm	2.14	1.66
0.075 mm	1.10	0.97
0.038 mm	0.86	0.77
Mean particle size, ⁴ mm	8.36	6.92
Penn State Particle Separator, % DM retained		
>19.0 mm	7.19	5.69
19.0 to 8.0 mm	67.3	35.7
<8.0 mm	25.5	58.6
Fermentation		
pH	4.61	4.42
Acetic acid, % DM	4.42	2.17
Propionic acid, % DM	0.45	0.12
Butyric acid, % DM	0.26	<0.01
Lactic acid, % DM	9.01	8.40
Lactic acid:acetic acid	2.04	3.87
Ethanol, % DM	0.14	0.11
Ammonia, mM	8.09	6.04

¹iNDF = indigestible NDF.²Thirty-hour in vitro NDF digestibility.³Particle size distributions of silages were measured each period (n = 2).⁴Mean particle size calculated from particle size distribution determined by wet sieving.

contributed 50% of forage NDF. Diets also contained dry ground corn, soybean meal (48% CP), SoyPLUS (West Central Soy Cooperative, Ralston, IA), vitamin-mineral premix, and limestone (Table 3).

Data and Sample Collection

Throughout the experiment, cows were housed in tie-stalls and fed diets as TMR once daily (1130 h) at 110% of expected intake. The amount of feed offered and refused (orts) was weighed daily for each cow. Forage samples were collected twice weekly and analyzed to adjust diets to account for DM, NDF, and CP fluctuation. Samples of all dietary ingredients (0.5

kg) and orts (12.5%) were collected daily from d 11 to 14 during the preliminary period and d 11 to 15 during each experimental period. Samples were frozen immediately after collection at -20°C and combined into 1 composite sample per period before analysis.

Cows were moved to an exercise lot twice daily (0230 and 1300 h) before milking in a parlor (0400 and 1430 h). Milk yield was measured and milk was sampled at each milking on d 11 to 14 of the preliminary period and on d 11 to 15 of the experimental periods. Rumen-empty BW was measured by weighing the cow after evacuation of ruminal digesta on d 14 of the preliminary period and d 17 of each experimental period. Body condition score was determined on the same days by 4

Table 3. Ingredients and chemical composition of preliminary and treatment diets (as analyzed) containing either early- or late-cut alfalfa silage as the sole source of forage

Composition	Preliminary	Early	Late
Ingredient, % DM			
Alfalfa silage, early cut	27.0	53.4	—
Alfalfa silage, late cut	20.2	—	42.3
Dry ground corn	38.1	34.4	39.3
Soybean meal (48% CP)	6.94	—	13.0
SoyPLUS ¹	2.75	7.57	—
Vitamin-mineral mix ²	4.68	4.68	4.68
Limestone	0.39	—	0.78
Chemical composition			
DM, %	44.8	43.4	54.2
OM, % DM	94.1	91.8	92.8
NDF, % DM	26.8	27.1	27.7
% forage NDF	22.0	21.8	22.4
% NDF from forage	82.0	80.4	81.1
iNDF, ³ % DM	NA ⁴	17.5	19.9
iNDF, % of NDF	NA	64.6	71.8
CP, % DM	18.4	20.2	19.0
Starch, % DM	32.1	28.2	32.1

¹West Central Soy Cooperative (Ralston, IA).

²Vitamin-mineral mix contained (DM basis) 17.1% sodium bicarbonate, 3.9% dicalcium phosphate, 2.6% magnesium oxide, 1.9% salt, 1.9% trace mineral premix, 0.4% vitamin A, 0.4% vitamin D, 0.2% vitamin E, and 71.6% dry ground corn as a carrier.

³iNDF = indigestible NDF.

⁴NA = no analysis for preliminary diet.

trained investigators blinded to treatments (Wildman et al., 1982; 5-point scale where 1 = thin and 5 = fat).

Duodenal samples (900 mL), fecal samples (500 g), rumen fluid, and particulate samples for microbial isolation (400 g), and rumen fluid samples for pH and concentrations of VFA, lactate, and ammonia (100 mL) were collected every 15 h from d 11 to 15 of each experimental period so that 8 samples were taken for each cow in each period, representing every 3 h of a 24-h period to account for diurnal variation. Rumen fluid and particulate matter for microbial isolation were collected from the reticulum, near the reticular-omasal orifice, transported to the laboratory, and processed. Rumen fluid for pH, VFA, lactate, and ammonia was obtained by combining digesta from 5 different sites in the rumen and straining it through nylon mesh (~1 mm pore size); fluid pH was recorded immediately. Samples were stored at -20°C .

Ruminal contents were evacuated manually through the ruminal cannula 4.5 h after feeding at the beginning of d 16 (1600 h), and 41.5 h later, 2 h before feeding at the end of d 17 (0930 h) for each experimental period. Total rumen content mass and volume were determined. To ensure accurate sampling, every tenth handful of digesta (10%) was separated for a subsample throughout evacuation. This subsample was squeezed into primarily solid and liquid phases. Both phases were weighed and sampled (350 mL) for determination of nutrient pool size. All samples were stored at -20°C .

Sample Analysis and Calculations

Milk yields recorded at each milking were summed for a daily total, which were averaged for each period. Milk samples were analyzed for fat, true protein, lactose, and SNF with infrared spectroscopy by Michigan DHIA (East Lansing). Yields of 3.5% FCM and milk components were calculated using milk yield and component concentrations for each milking, summed for a daily total, and averaged for each period.

Forage samples were combined to 1 composite sample per forage per period. Particle size distribution was determined using the Penn State Particle Separator containing 2 sieves (19 and 8 mm) and a pan (Lammers et al., 1996). In addition, samples were wet sieved manually and sequentially through screens with the following aperture sizes: 19.0, 9.50, 4.75, 2.36, 1.18, 0.600, 0.300, 0.150, 0.075, and 0.038 mm. The fraction of DM retained on the screens from wet sieving was used to calculate mean particle size.

Diet ingredients, Orts, and feces were lyophilized (Tri-Philizer MP; FTS Systems Inc., Stone Ridge, NY). All dried samples were ground with a Wiley mill (1-mm screen; Arthur H. Thomas Co., Philadelphia, PA). Dried, ground fecal samples were combined on an equal DM basis into 1 sample per cow per period. Frozen duodenal samples for each cow period ($n = 8$) were chopped finely using a commercial food processor (84142 Food cutter; Hobart Manufacturing Co., Troy,

OH) and subsampled in the frozen state to obtain representative samples. These duodenal subsamples and the 350 mL of ruminal solid and liquid samples were lyophilized and ground as described above. Dried ruminal solid and liquid samples were recombined according to the original ratio of solid and liquid DM.

Samples were analyzed for ash, NDF, indigestible NDF (**iNDF**), ADF, acid detergent sulfuric acid lignin (ADL), CP, and starch content. Ash concentration was determined after 5 h combustion at 500°C in a muffle furnace. Concentrations of NDF were determined according to Mertens (2002) and ADF and ADL according to Goering and Van Soest (1970). Indigestible NDF was estimated as NDF residue after 240-h in vitro fermentation (Goering and Van Soest, 1970); flasks were reinoculated at 120 h to ensure a viable microbial population. Forage NDF digestibility was determined by 30-h in vitro fermentation (Goering and Van Soest, 1970). Ruminal fluid for the in vitro incubations was collected from a nonpregnant dry cow fed dry hay only. The fraction of potentially digestible NDF (**pdNDF**) was calculated by difference ($1.00 - \text{iNDF}$). Crude protein was analyzed according to Hach et al. (1987). Starch content was measured by an enzymatic method (Karkalas, 1985) after samples were gelatinized with sodium hydroxide. Glucose concentration was measured using a glucose oxidase method (Glucose kit #510; Sigma Chemical Co., St. Louis, MO), and absorbance was determined with a microplate reader (SpectraMax 190; Molecular Devices Corp., Sunnyvale, CA). Concentrations of all nutrients except DM were expressed as percentages of DM determined by drying at 105°C in forced-air oven for more than 8 h.

Duodenal digesta were analyzed for purines and ammonia to estimate microbial N (**MN**) flow and nonammonia, nonmicrobial N (**NANMN**) flow to the duodenum. Purine concentration was used as a microbial marker, and purine-to-MN ratio was estimated by analysis of microbial pellets obtained by differential centrifugation of the rumen fluid and particulate samples collected near the reticulum. Rumen fluid and particulate matter were blended, strained through nylon mesh, and the liquid portion was centrifuged at $500 \times g$ for 15 min. The supernatant was centrifuged at $18,000 \times g$ for 15 min, and the pellet was washed with 0.9% NaCl solution and centrifuged again at $18,000 \times g$ for 15 min, resuspended in water, and lyophilized. Total purines were measured by spectrophotometer (Beckman Instruments Inc., Fullerton, CA) at 260 nm according to Zinn and Owens (1986). Ammonia concentration was determined for centrifuged duodenal and rumen fluid samples according to Broderick and Kang (1980). Rumen fluid was also analyzed for concentrations of

major VFA and lactate by HPLC (Waters Corp., Milford, MA) according to Oba and Allen (2003a).

Dry matter and nutrient intakes were calculated using the composition of feed offered and refused. Ruminal pool sizes (kg) of OM, NDF, iNDF, pdNDF, starch, MN, and NANMN were determined by multiplying the concentration of each component in rumen samples by the ruminal digesta DM mass (kg). Duodenal flows (kg/d) of DM, OM, total NDF, pdNDF, starch, MN, NANMN, and ammonia N were determined using iNDF as a flow marker; iNDF intake (kg/d) was multiplied by the ratio between the component and iNDF in duodenal digesta. Duodenal flow of microbial OM was determined using the purines-to-OM ratio (Oba and Allen, 2003a), and true ruminally digested OM was calculated by subtracting duodenal flow of nonmicrobial OM from OM intake. Indigestible NDF was used as an internal marker to estimate nutrient digestibility in the rumen and in the total tract (Cochran et al., 1986). Turnover rate in the rumen, passage rate from the rumen, and ruminal digestion rate of each component was calculated by the following equations:

$$\text{Turnover rate (\%/h)} = 100 \times (\text{intake of component} / \text{ruminal pool of component}) / 24;$$

$$\text{Passage rate (\%/h)} = 100 \times (\text{duodenal flow of component} / \text{ruminal pool of component}) / 24;$$

$$\text{Digestion rate (\%/h)} = \text{turnover rate in the rumen (\%/h)} - \text{passage rate from the rumen (\%/h)}.$$

Statistical Analysis

All data were analyzed by using the fit model procedure of JMP (version 8; SAS Institute Inc., Cary, NC). To determine differences between treatments and evaluate interactions of treatment with DMI, where pDMI (calculated as the mean of DMI values on d 11 to 14 of the 14-d preliminary period) was used as the covariate for treatment responses, data were analyzed according to the following model: $Y_{ijk} = \mu + C_i + P_j + T_k + PT_{jk} + \text{pDMI} + T_k \text{pDMI} + \text{pDMI}^2 + T_k \text{pDMI}^2 + e_{ijk}$, where Y_{ijk} is the dependent variable, μ is the overall mean, C_i is the random effect of cow ($i = 1$ to 16), P_j is the fixed effect of period ($j = 1$ to 2), T_k is the fixed effect of treatment ($k = 1$ to 2), PT_{jk} is the interaction of period and treatment, pDMI is the linear effect of pDMI, $T_k \text{pDMI}$ is the interaction of treatment and pDMI (linear), pDMI^2 is the quadratic effect of pDMI, $T_k \text{pDMI}^2$ is the interaction of treatment and pDMI (quadratic), and e_{ijk} is the residual error. Statistical significance for $T_k \text{pDMI}$

and $T_{kp}DMI^2$ indicated that treatment differences were related to pDMI. Covariate and interaction terms were removed stepwise from the model if $P > 0.20$. Treatment effects and their interaction (linear and quadratic relationships) were declared significant at $P \leq 0.05$ and $P \leq 0.10$, respectively. Tendencies for treatment effects and their interactions were declared at $P \leq 0.10$ and $P \leq 0.15$, respectively.

RESULTS AND DISCUSSION

Comparison of Forages and Diets

Chemical analyses and physical characteristics of AL silages are listed in Table 2. As expected, delaying harvest of AL decreased the concentration of CP (23.7 vs. 18.1%) and increased concentrations of total NDF (40.8 vs. 53.1%), iNDF (27.7 vs. 39.3%), and ADL (8.41 vs. 11.9%). Indigestible NDF, expressed as a percent of NDF, was lower for early-cut AL than for late-cut AL (67.8 vs. 74.1% of NDF), but was unusually high for both silages. In vitro NDF digestibility (30-h) was 4.4 percentage units higher for early-cut AL than for late-cut AL (28.8 vs. 24.4%). Early-cut AL had lower DM concentration, higher pH, and contained more lactic and acetic acids than late-cut AL. Based on wet sieving, early-cut AL had a greater mean particle size than late-cut AL (8.36 vs. 6.92 mm). Additionally, early-cut AL contained a larger proportion of particles between 8 and 19 mm (67.3 vs. 35.7%; middle sieve) and a smaller proportion of particles less than 8 mm (25.5 vs. 58.6%; bottom pan) than late-cut AL when sieved with the Penn State Particle Separator. Although AL was chopped to the same theoretical length of cut for both silages, the smaller particle size for late-cut AL may be due to greater fragility from increased maturity or greater DM concentration, or both.

Diet ingredients and chemical composition are shown in Table 3. The preliminary diet contained more early-cut AL than late-cut AL so each forage supplied similar concentrations of forage NDF. Because treatment diets were formulated to contain similar forage NDF concentrations, forage:concentrate ratios were different between diets with ratios of 53:47 and 42:58 for EC and LC, respectively. Besides forage source, differences in diets included source and concentration of protein supplement and concentration of corn grain, which were both lower for EC than LC, to account for differences between the AL silages. Additionally, limestone was added to LC to compensate for lower Ca concentration in more mature AL silage. The chemical composition of each ration was mathematically calculated according to the proportion of each feed ingredient in the diet and its respective analytical values, and was similar between

rations for OM, total NDF, and CP concentrations. Based on the calculated diet composition, both diets contained lower NDF and higher CP concentrations than the formulated targets but were similar for EC and LC and NDF concentration was above NRC (2001) minimum requirements. Starch concentration was lower for EC because of more forage and less concentrate in the diet for EC compared with LC. Differences in DM and iNDF concentrations in diets, which were lower for EC than LC, were reflective of the concentrations in the forages, which were lower for early-cut AL than late-cut AL. In both diets, forage NDF provided over 80% of the total diet NDF.

Effects of Legume Maturity and pDMI

The EC diet increased DMI (28.6 vs. 26.8 kg/d, $P = 0.003$; Table 4) and total NDF intake (7.63 vs. 7.30 kg/d, $P = 0.03$; Table 5) compared with the LC diet, but these responses were not related to level of intake (interaction $P \geq 0.53$). We expected LC to be more filling than EC, causing greater rumen distention and potentially limiting DMI, particularly in cows with high DMI for which ruminal distention is more likely to limit feed intake (Allen, 1996). Oba and Allen (1999) and Voelker et al. (2002) found that DMI responses to a more filling diet varied by production level, which is generally correlated with DMI (NRC, 2001), where DMI was increasingly limited by high-fill diets compared with low-fill diets as milk yield increased. As fill became more limiting, we expected ruminal passage rate of iNDF to increase more for EC than for LC. However, response of passage rate of iNDF to treatment was not related to pDMI (interaction $P \geq 0.37$), but EC increased ruminal passage rate of iNDF compared with LC (3.86 vs. 3.57%/h, $P = 0.01$; Table 6).

The lower concentration and faster rate of passage of iNDF for EC is reflected in the lower rumen pool sizes of iNDF (5.49 vs. 6.24 kg, $P < 0.001$) and NDF (6.23 vs. 6.96 kg, $P < 0.001$) for EC compared with LC (Table 7), which increased at a slower rate for EC than LC as pDMI increased (interaction $P \leq 0.10$; Figure 1). It is also partially responsible for lower rumen pool sizes of OM (10.3 vs. 11.4 kg, $P = 0.006$) and DM (11.3 vs. 12.4, $P = 0.007$) for EC compared with LC (Table 7), which tended to increase at a slower rate for EC than LC as pDMI increased (interaction $P \leq 0.14$). Furthermore, EC decreased rumen digesta wet weight (76.9 vs. 80.8 kg, $P = 0.04$) and volume (91.3 vs. 97.8 L, $P = 0.03$) compared with LC (Table 7). The faster rate of passage of iNDF and lower rumen pools for EC are consistent with the faster rates of ruminal turnover of NDF, OM, and DM (5.25 vs. 4.45, 10.8 vs. 9.15, and 10.7 vs. 9.07%/h, respectively, $P < 0.001$) for EC com-

Table 4. Milk production and composition, feed intake, and BW change of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Item	Treatment LSM			<i>P</i> -value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
Yield, kg/d									
Milk	42.5	41.7	2.6	0.14	NS ²	0.001	NS	0.09	NS
FCM (3.5%)	44.0	43.0	2.4	0.24	NS	0.003	NS	0.13	NS
Milk fat	1.59	1.55	0.08	0.33	NS	0.01	NS	0.19	NS
Milk protein	1.31	1.30	0.05	0.62	NS	<0.001	NS	0.04	NS
Milk lactose	2.06	2.03	0.15	0.30	NS	0.001	NS	0.08	NS
SNF	3.76	3.71	0.22	0.40	NS	<0.001	NS	0.06	NS
Milk composition, %									
Fat	3.78	3.74	0.12	0.49	NS	<0.001	0.09	0.06	NS
Protein	3.19	3.23	0.07	0.18	NS	0.03	NS	NS	NS
Lactose	4.82	4.85	0.08	0.30	NS	0.002	NS	0.05	NS
SNF	8.76	8.83	0.07	0.06	NS	NS	NS	NS	NS
DMI, kg/d	28.6	26.8	0.7	0.003	0.02	<0.001	NS	0.14	NS
3.5% FCM/DMI	1.45	1.51	0.05	0.01	0.02	0.03	NS	0.17	NS
BW change, kg/17 d	4.70	4.61	3.03	0.99	NS	0.02	NS	0.03	NS
BCS change/17 d	0.065	0.067	0.044	0.98	0.11	<0.001	0.72	0.001	0.08

¹*P*-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with *P* > 0.20; term was removed from the statistical model.

pared with LC (Table 6). The decreased feed intake of LC was accompanied by increased size of rumen pools, suggesting rumen fill as a constraint limiting DMI for cows consuming LC is possible, but no evidence existed in this experiment that ruminal distention is more likely

to limit feed intake for cows with high intake compared with cows with low intake, as we were unable to detect a treatment by pDMI interaction for DMI.

Although EC increased DMI compared with LC, forage maturity and its interaction with pDMI did not

Table 5. Neutral detergent fiber digestion of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Item	Treatment LSM			<i>P</i> -value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
NDF									
Intake, kg/d	7.63	7.30	0.20	0.03	0.15	<0.001	NS ²	0.14	NS
Ruminal digestion									
kg/d	2.63	2.15	0.13	0.003	NS	0.07	NS	NS	NS
%	35.7	30.2	1.5	0.02	NS	NS	NS	NS	NS
Passage to duodenum, kg/d	4.97	5.12	0.17	0.48	0.03	<0.001	NS	0.08	NS
Postruminal digestion, kg/d	−0.50	−0.74	0.14	0.19	NS	NS	NS	NS	NS
Total-tract digestion									
kg/d	2.13	1.40	0.09	<0.001	NS	0.02	NS	NS	NS
%	28.5	19.5	1.0	<0.001	NS	NS	NS	NS	NS
Potentially digestible NDF									
Intake, kg/d	2.71	2.06	0.07	<0.001	NS	<0.001	NS	0.16	NS
Ruminal digestion									
kg/d	2.63	2.15	0.12	0.003	NS	0.07	NS	NS	NS
%	99.9	107.8	5.1	0.29	NS	NS	NS	NS	NS
Passage to duodenum, kg/d	0.01	−0.15	0.11	0.29	NS	NS	NS	NS	NS
Postruminal digestion, kg/d	−0.50	−0.74	0.14	0.19	NS	NS	NS	NS	NS
Total-tract digestion									
kg/d	2.13	1.40	0.09	<0.001	NS	0.02	NS	NS	NS
%	80.0	69.2	3.3	0.007	NS	NS	NS	NS	NS
Indigestible NDF									
Intake, kg/d	4.92	5.24	0.14	0.008	0.12	<0.001	NS	0.14	NS

¹*P*-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with *P* > 0.20; term was removed from the statistical model.

Table 6. Rumen kinetics of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Item	Treatment LSM			P-value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
Ruminal turnover rate, %/h									
DM	10.7	9.07	0.31	<0.001	0.002	0.07	NS ²	0.10	NS
OM	10.8	9.15	0.33	<0.001	0.002	0.09	NS	0.11	NS
NDF	5.25	4.45	0.15	<0.001	0.04	0.05	NS	0.04	NS
pdNDF ³	17.2	14.1	1.5	0.15	NS	0.80	NS	0.19	NS
Starch	36.5	33.1	2.2	0.21	0.02	NS	NS	NS	NS
Ruminal turnover time, h									
DM	9.56	11.1	0.33	<0.001	0.004	0.03	NS	0.08	NS
OM	9.54	11.0	0.34	0.002	0.003	0.04	NS	0.09	NS
NDF	19.3	22.7	0.71	<0.001	0.10	0.03	NS	0.03	NS
pdNDF	6.70	8.90	0.66	0.05	NS	NS	NS	NS	NS
iNDF ⁴	26.6	28.2	1.1	0.11	0.09	0.03	NS	0.07	NS
Starch	2.93	3.19	0.15	0.21	0.005	NS	NS	NS	NS
Ruminal passage rate, %/h									
pdNDF	-0.06	-0.91	0.77	0.42	NS	NS	NS	NS	NS
iNDF	3.86	3.57	0.13	0.01	0.05	0.05	NS	0.07	NS
Starch	19.7	18.4	2.2	0.57	NS	NS	NS	NS	NS
Ruminal digestion rate, %/h									
pdNDF	16.1	13.9	1.5	0.37	NS	NS	NS	NS	NS
Starch	17.1	13.0	1.8	0.03	0.01	0.57	0.06	0.58	0.13

¹P-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with $P > 0.20$; term was removed from the statistical model.

³pdNDF = potentially digestible NDF.

⁴iNDF = indigestible NDF.

affect milk, milk component, or FCM yields (Table 4). Therefore, EC decreased efficiency of milk production (FCM/DMI) compared with LC (1.45 vs. 1.51, $P = 0.01$), as cows consuming EC ate more feed than cows consuming LC to produce similar yields of FCM. Response of milk fat concentration to treatment was related to pDMI (interaction $P = 0.09$; Table 4) such that EC decreased milk fat concentration more per

kilogram of pDMI increase than did LC. However, this effect was small and milk fat yield was not affected. The aforementioned results are likely because of the greater concentration of supplemental concentrate in the diet for LC.

The different forage:concentrate ratios in the diets were necessary to account for changes in chemical composition of AL with increasing maturity and maintain

Table 7. Rumen pools of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Item	Treatment LSM			P-value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
Wet weight, kg	76.9	80.8	2.2	0.04	NS ²	0.08	NS	NS	NS
Volume, L	91.3	97.8	3.0	0.03	NS	0.07	NS	NS	NS
Density, kg/L	0.84	0.83	0.01	0.29	NS	NS	NS	NS	NS
Rumen pool, kg									
DM	11.3	12.4	0.3	0.007	NS	0.07	0.13	NS	NS
OM	10.3	11.4	0.3	0.006	NS	0.08	0.14	NS	NS
NDF	6.23	6.96	0.17	<0.001	NS	0.06	0.10	NS	NS
pdNDF ³	0.75	0.73	0.05	0.83	NS	0.007	NS	NS	NS
iNDF ⁴	5.49	6.24	0.18	<0.001	NS	0.18	0.08	NS	NS
Starch	0.964	1.13	0.051	0.02	0.06	0.20	NS	NS	NS

¹P-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with $P > 0.20$; term was removed from the statistical model.

³pdNDF = potentially digestible NDF.

⁴iNDF = indigestible NDF.

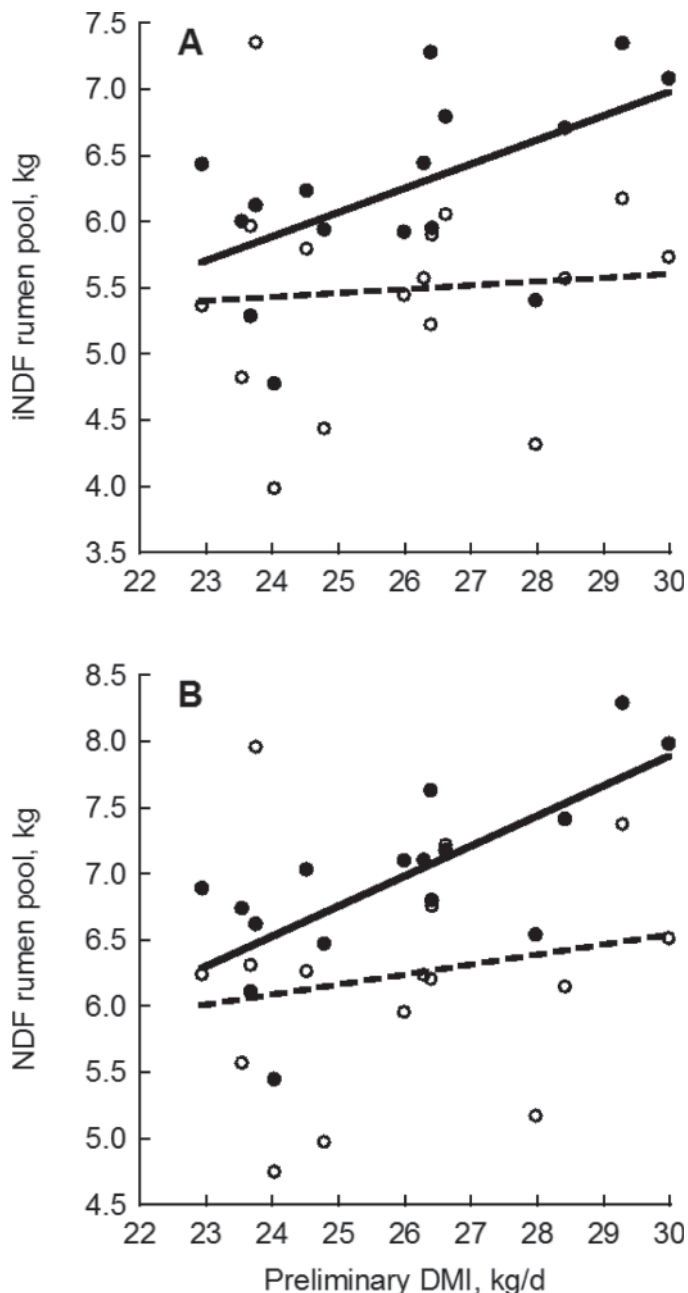


Figure 1. Interaction of early (open circles, dashed line) and late (closed circles, solid line) alfalfa maturity with preliminary DMI for A) indigestible NDF (iNDF) rumen pool (interaction: $P = 0.08$ linear; early: $P = 0.78$, $R^2 = 0.01$; late: $P = 0.03$, $R^2 = 0.30$) and B) NDF rumen pool (interaction: $P = 0.10$ linear; early: $P = 0.48$, $R^2 = 0.04$; late: $P = 0.002$, $R^2 = 0.51$). The preliminary DMI on the x-axis are the mean DMI of individual cows during the final 4 d of the preliminary period when all cows were fed a common diet. The best-fit lines are drawn to demonstrate the significant interaction even if the individual relationships are not significant.

the same concentration of forage NDF in the diets. As a result, EC decreased starch intake (8.16 vs. 8.74 kg/d, $P = 0.007$; Table 8), decreased ruminal propionate (29.5 vs. 31.0 mM, $P = 0.04$) and butyrate (16.1 vs. 18.1

mM, $P < 0.001$) concentrations, and increased ruminal acetate concentration (86.3 vs. 81.9 mM, $P = 0.005$) compared with LC (Table 9). The EC diet increased the acetate-to-propionate ratio compared with the LC diet (2.94 vs. 2.65, $P < 0.001$), and the difference in ruminal acetate concentration between EC and LC was greatest for cows with low pDMI (interaction $P = 0.09$, Figure 2). These results are reflective of the diets.

The EC diet increased mean ruminal pH (6.36 vs. 6.29, $P = 0.04$; Table 9) compared with the LC diet, which was expected because of lower starch intake for EC compared with LC, but the reason for the difference in pH was not clear based on our results. Apparent ruminal digestion of OM was greater for EC than LC (9.76 vs. 7.93 kg/d, $P < 0.001$; Table 10), which would suggest greater production of VFA and lower pH for EC, but total VFA concentration in the rumen was similar for EC and LC ($P = 0.48$; Table 9), whereas pH was higher for EC than LC. Additionally, rumen content mass was smaller for EC than LC (Table 7) suggesting less buffering capacity of digesta. The ruminal pH difference observed might be due to chewing activity and buffering through saliva secretion but these responses were not measured. In an experiment of similar design evaluating grass maturity in which chewing behavior was measured, cows spent less time ruminating per day when consuming early-cut orchardgrass compared with late-cut orchardgrass (K. L. Kammes and M. S. Allen, unpublished data). However, it is not known if differences in ruminating time because of maturity would be the same for legumes and grasses.

The EC diet increased pdNDF intake (2.71 vs. 2.06 kg/d, $P < 0.001$) and decreased iNDF intake (4.92 vs. 5.24 kg/d, $P = 0.008$) compared with the LC diet (Table 5) because of differences in chemical composition of forages. Despite the differences in the proportion of iNDF and pdNDF within the total NDF of the forage, both treatments resulted in almost complete digestion of pdNDF in the rumen, leading to essentially no passage of pdNDF to the duodenum (Table 5), and therefore, ruminal passage rate of pdNDF was negligible (Table 6). Ruminal digestibility of pdNDF in the current experiment was unusually high because the pdNDF fraction was abnormally low. The EC diet increased ruminal (35.7 vs. 30.2%, $P = 0.02$) and total-tract (28.5 vs. 19.5%, $P < 0.001$) digestibility of NDF compared with the LC diet (Table 5), which was because of a greater proportion of pdNDF and lower proportion of iNDF in NDF for EC compared with LC (Table 2).

Total-tract digestibilities of NDF (and pdNDF) were lower than ruminal digestibility because negative postruminal digestibilities were calculated for NDF (and pdNDF) in the present experiment. This was due to a net gain of fiber from the duodenum to the

Table 8. Starch digestion of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Starch	Treatment LSM			P-value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
Intake, kg/d	8.16	8.74	0.23	0.007	0.04	0.001	NS ²	0.18	NS
Apparent ruminal digestion kg/d	3.87	3.74	0.32	0.67	0.11	NS	NS	NS	NS
%	48.2	44.6	3.8	0.29	NS	NS	NS	NS	NS
True ruminal digestion kg/d	4.01	3.90	0.32	0.73	0.12	NS	NS	NS	NS
%	49.9	46.5	3.8	0.31	NS	NS	NS	NS	NS
Passage to duodenum, kg/d	4.07	4.78	0.35	0.04	NS	0.13	NS	NS	NS
Apparent postruminal digestion kg/d	3.46	4.01	0.34	0.13	NS	0.20	NS	NS	NS
% of intake	44.2	46.5	3.8	0.53	NS	NS	NS	NS	NS
% of duodenal passage	84.5	82.9	1.3	0.36	NS	NS	NS	NS	NS
Apparent total-tract digestion kg/d	7.33	7.74	0.15	0.02	0.03	0.002	NS	NS	NS
%	92.3	91.1	0.5	0.05	NS	0.11	NS	NS	NS

¹P-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with $P > 0.20$; term was removed from the statistical model.

feces, which has previously been reported with both the gutter-type T duodenal cannula (Huhtanen and Jaakkola, 1993; Poore et al., 1993), which is the type used in this study, and closed T-type duodenal cannula (Stensig and Robinson, 1997). The underestimation of duodenal NDF flow or duodenal iNDF:NDF ratio using iNDF as a marker creates inaccuracies of estimated flow of duodenal fiber and postruminal digestibility. These errors may be related to unrepresentative digesta sampling due to differential separation of fluid and particles relative to the true material flowing out of the duodenum or analytical problems in fiber determination

of duodenal samples, possibly due to a component in the duodenal digesta that interferes with the analysis. Although absolute values are not biologically reasonable, relative comparisons between treatments within the same experiment are useful.

The EC diet increased true ruminal OM digestibility (52.5 vs. 46.4%, $P = 0.002$) and digestion (13.4 vs. 11.1 kg/d, $P < 0.001$) compared with the LC diet (Table 10). Because postruminal OM digestibility and digestion were similar for EC and LC, EC also increased total-tract OM digestibility (62.8 vs. 58.3%, $P < 0.001$) and digestion (16.1 vs. 14.1 kg/d, $P < 0.001$) compared

Table 9. Ruminal VFA concentrations and pH of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Item	Treatment LSM			P-value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
Total VFA, mM	141	139	2.0	0.48	NS ²	NS	NS	NS	NS
Acetate	86.3	81.9	1.0	0.005	NS	0.74	0.09	NS	NS
Propionate	29.5	31.0	0.6	0.04	NS	NS	NS	NS	NS
Butyrate	16.1	18.1	0.4	<0.001	0.06	NS	NS	NS	NS
Lactate	1.79	1.37	0.33	0.37	NS	0.31	NS	0.06	NS
Isobutyrate	1.61	1.54	0.02	0.02	NS	0.11	NS	NS	NS
Valerate	2.68	2.47	0.07	0.05	NS	0.13	0.95	0.60	0.02
Isovalerate	2.82	2.51	0.09	0.003	NS	0.07	0.04	0.02	0.002
Branched-chain VFA ³	4.41	4.03	0.10	0.005	NS	0.02	0.09	0.01	0.01
Acetate:propionate	2.94	2.65	0.05	<0.001	NS	NS	NS	NS	NS
Ruminal pH	6.36	6.29	0.02	0.04	NS	NS	NS	NS	NS

¹P-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with $P > 0.20$; term was removed from the statistical model.

³Branched-chain VFA include isobutyrate and isovalerate.

Table 10. Dry matter and OM digestion of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Item	Treatment LSM			P-value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
DM									
Intake, kg/d	28.6	26.8	0.7	0.003	0.02	<0.001	NS ²	0.14	NS
Apparent total-tract digestion									
kg/d	17.4	15.1	0.4	<0.001	0.005	0.003	NS	NS	NS
%	62.5	58.0	0.5	<0.001	0.05	0.60	0.12	NS	NS
OM									
Intake, kg/d	26.2	24.8	0.7	0.009	0.02	<0.001	NS	0.14	NS
Apparent ruminal digestion									
kg/d	9.76	7.93	0.41	<0.001	0.009	NS	NS	NS	NS
%	37.9	33.2	1.6	0.007	0.14	NS	NS	NS	NS
True ruminal digestion									
kg/d	13.4	11.1	0.5	<0.001	0.03	NS	NS	NS	NS
%	52.5	46.4	1.7	0.002	NS	NS	NS	NS	NS
Passage to duodenum, kg/d	16.5	17.0	0.7	0.40	NS	0.001	NS	0.10	NS
Apparent postruminal digestion									
kg/d	6.30	6.12	0.44	0.73	NS	0.05	NS	NS	NS
% of intake	24.8	25.1	1.7	0.90	NS	NS	NS	NS	NS
Apparent total-tract digestion									
kg/d	16.1	14.1	0.3	<0.001	0.003	0.003	NS	NS	NS
%	62.8	58.3	0.5	<0.001	0.03	NS	NS	NS	NS

¹P-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with $P > 0.20$; term was removed from the statistical model.

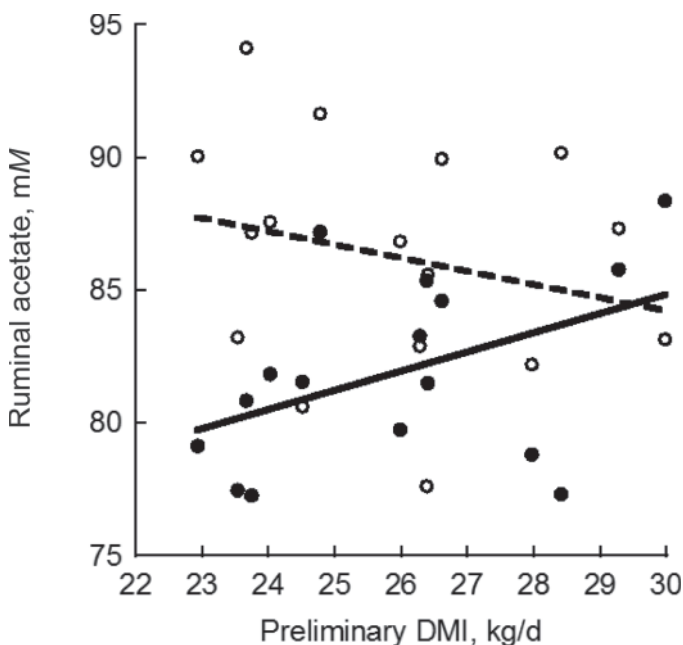


Figure 2. Interaction of early (open circles, dashed line) and late (closed circles, solid line) alfalfa maturity with preliminary DMI for ruminal acetate concentration (interaction: $P = 0.09$ linear; early: $P = 0.36$, $R^2 = 0.06$; late: $P = 0.09$, $R^2 = 0.19$). The preliminary DMI on the x-axis are the mean DMI of individual cows during the final 4 d of the preliminary period when all cows were fed a common diet. The best-fit lines are drawn to demonstrate the significant interaction even if the individual relationships are not significant.

with LC. This resulted in greater total-tract DM digestibility (62.5 vs. 58.0%, $P < 0.001$) and digestion (17.4 vs. 15.1 kg/d, $P < 0.001$) for EC than for LC (Table 10).

Despite less supplementation with soybean products in EC compared with LC, the high CP concentration in the early-cut AL silage resulted in greater N intake for EC than LC (918 vs. 812 g/d, $P < 0.001$; Table 11). The EC diet increased ruminal ammonia concentration compared with the LC diet (22.5 vs. 20.4 mg/dL, $P < 0.001$; Table 11). Early-cut AL had higher ammonia concentration than late-cut AL (8.09 vs. 6.04 mM; Table 2), which may be one source for the greater ammonia concentration observed for cows consuming EC. Another source might be protein degradation in the rumen, as indicated by greater concentrations of isovalerate (2.82 vs. 2.51 mM, $P = 0.003$) and branched-chain VFA (4.41 vs. 4.03 mM, $P = 0.005$) for EC than LC (Table 9). This is consistent with the faster rate of ruminal starch digestion for EC compared with LC (Table 6); early-cut AL might promote greater numbers or activity of starch-digesting bacteria in the rumen of cows consuming EC. Because some starch-digesting bacteria in the rumen (e.g., *Streptococcus bovis*) also have high proteolytic activity (Russell et al., 1981), resulting in deamination of AA and production of ammonia, this could contribute to the increased ammonia concentration for EC.

Table 11. Nitrogen metabolism of cows fed treatment diets containing either early- or late-cut alfalfa silage as the sole source of forage

Item	Treatment LSM			P-value ¹					
	Early	Late	SE	Trt	Trt × period	pDMI	Trt × pDMI	pDMI × pDMI	Trt × pDMI × pDMI
N intake, g/d	918	812	23	<0.001	0.07	<0.001	NS ²	0.14	NS
Ruminal ammonia, mg/dL	22.5	20.4	0.6	<0.001	NS	0.58	0.14	NS	NS
Flow to duodenum									
Ammonia N, g/d	17.8	16.9	0.9	0.31	NS	0.004	NS	NS	NS
NAN									
g/d	727	715	27	0.49	NS	0.001	NS	0.07	NS
% of N intake	79.5	88.3	2.4	<0.001	NS	0.13	NS	0.17	NS
NANMN ³									
g/d	292	321	19	0.14	0.16	0.03	0.17	NS	NS
% of N intake	32.5	41.1	2.3	0.008	NS	NS	NS	NS	NS
% of duodenal NAN	41.8	47.6	2.5	0.09	NS	0.31	0.17	NS	NS
Microbial N									
g/d	397	357	19	0.13	NS	0.11	NS	NS	NS
% of duodenal NAN	58.2	52.4	2.5	0.09	NS	0.31	0.17	NS	NS
g/kg of TRDOM ⁴	30.2	32.2	1.4	0.27	0.03	0.54	0.14	NS	NS
NAN apparent postruminal digestion									
g/d	405	394	22	0.53	0.15	0.01	NS	0.15	NS
% of N intake	42.6	46.8	1.8	0.06	NS	NS	NS	NS	NS
% of duodenal passage	55.1	54.2	1.1	0.56	0.10	NS	NS	NS	NS
N apparent total-tract digestion									
g/d	586	480	13	<0.001	0.02	0.005	NS	NS	NS
%	64.7	60.0	0.7	<0.001	0.02	0.05	0.20	0.08	NS

¹P-values for treatment (Trt), Trt by period interaction (Trt × period), preliminary DMI (pDMI), Trt by pDMI interaction (Trt × pDMI), quadratic effect of pDMI (pDMI × pDMI), and Trt by quadratic effect of pDMI (Trt × pDMI × pDMI).

²Nonsignificant, with $P > 0.20$; term was removed from the statistical model.

³NANMN = nonammonia, nonmicrobial nitrogen.

⁴TRDOM = true ruminally digested OM.

Effects of pDMI on Ruminal Passage Rates

Experimental data on rates of passage from the rumen, particularly for individual feed fractions, are scarce. Given the effect of passage on ruminal digestion and microbial growth, quantitative knowledge of rates of nutrient passage from the rumen is needed to better understand nutrient availability in ruminants and improve nutrition models. Furthermore, because passage rates from the rumen generally increase with increased intake, measurements of ruminal passage rates of nutrients over a wide range of DMI are necessary. We measured the effects of DMI on rates of passage of feed fractions from the rumen using the pool and flux method (Robinson et al., 1987).

Rate of ruminal passage for iNDF was related to level of intake independent of treatment ($P = 0.07$), which increased quadratically as pDMI increased with a greater increase for cows with lower pDMI (Figure 3). This is in contrast to results in the previously mentioned experiment of similar design evaluating effects of grass maturity in which cows in the middle of the pDMI range had the slowest rate of iNDF passage (K. L. Kammes and M. S. Allen, unpublished data). As previously discussed, ruminal digestion of pdNDF was

almost complete for all cows regardless of treatment and varying levels of pDMI; therefore, rate of ruminal passage for pdNDF was negligible and not related to level of intake.

The level of intake had no effect on the rate of starch passage, but it tended to influence rate of ruminal starch digestion (Table 6). As pDMI increased, EC slightly increased ruminal starch digestion rate and LC quadratically affected it (interaction $P = 0.13$, quadratic; Table 6). The largest difference in starch digestion rate between EC and LC was observed for cows in the middle of the pDMI range, such that EC had a greater starch digestion rate than LC (Figure 4A). However, this did not affect starch ruminal turnover rate (Table 6), digestibility (%), or digestion (kg/d; Table 8), which were similar for EC and LC. The EC diet decreased rumen pool of starch compared with the LC diet (0.964 vs. 1.13 kg, $P = 0.02$; Table 7) because of the lower intake and faster digestion rate of starch for EC but was not related to pDMI. The reduction in starch rate of digestion for LC might be related to increased liquid dilution rate (not measured) associated with greater increases in rumen pool sizes for LC compared with EC as intake increased, decreasing populations of starch-digesting microbes. Ruminal ammonia concentration

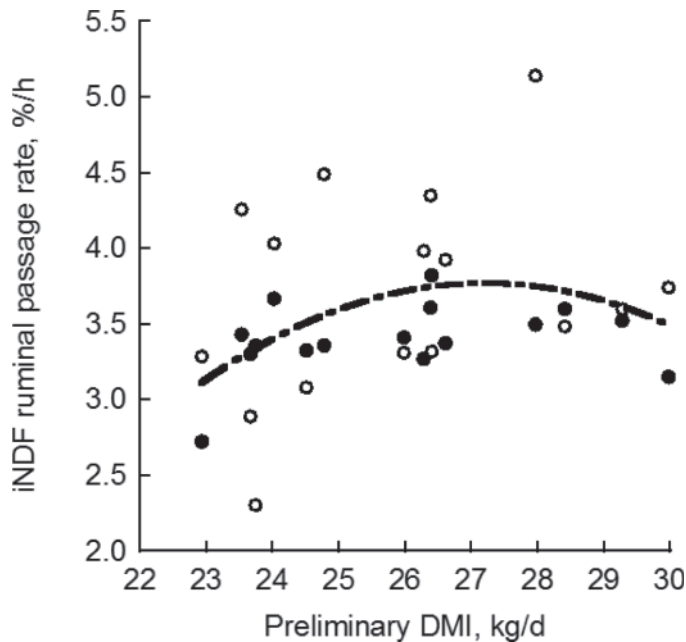


Figure 3. Relationship between indigestible NDF (iNDF) ruminal passage rate and preliminary DMI for early (open circles) and late (closed circles) alfalfa maturity ($P = 0.07$ quadratic; $R^2 = 0.15$). The preliminary DMI on the x-axis are the mean DMI of individual cows during the final 4 d of the preliminary period when all cows were fed a common diet.

tended to decrease with increased pDMI (interaction $P = 0.14$; Figure 4B) and the response of isovalerate (interaction $P = 0.002$, quadratic; Table 9) and branched-chain VFA (interaction $P = 0.01$, quadratic; Figure 4C) concentrations, which are derived from degradation of branched-chain AA, followed a pattern similar to starch rate of digestion for LC, but not EC. This is consistent with less proteolytic activity in the rumen for LC due to greater removal of ruminal microbes through passage or lysis.

Effects of Treatment and pDMI on N Flux and Microbial Efficiency

Results for N metabolism are shown in Table 11. The EC diet decreased NANMN flux (32.5 vs. 41.1% of N intake, $P = 0.008$) and NAN flux (79.5 vs. 88.3% of N intake, $P < 0.001$) to the duodenum as a percent of N intake and tended to increase MN flux (58.2 vs. 52.4% of duodenal NAN, $P = 0.09$) to the duodenum as a percent of duodenal NAN compared with the LC diet, but the amounts (g/d) of NANMN, NAN, and MN that passed from the rumen to the duodenum were not different ($P \geq 0.13$). However, fluxes of NANMN ($P = 0.03$), NAN ($P = 0.001$), and MN ($P = 0.11$) to the duodenum were positively related to pDMI indepen-

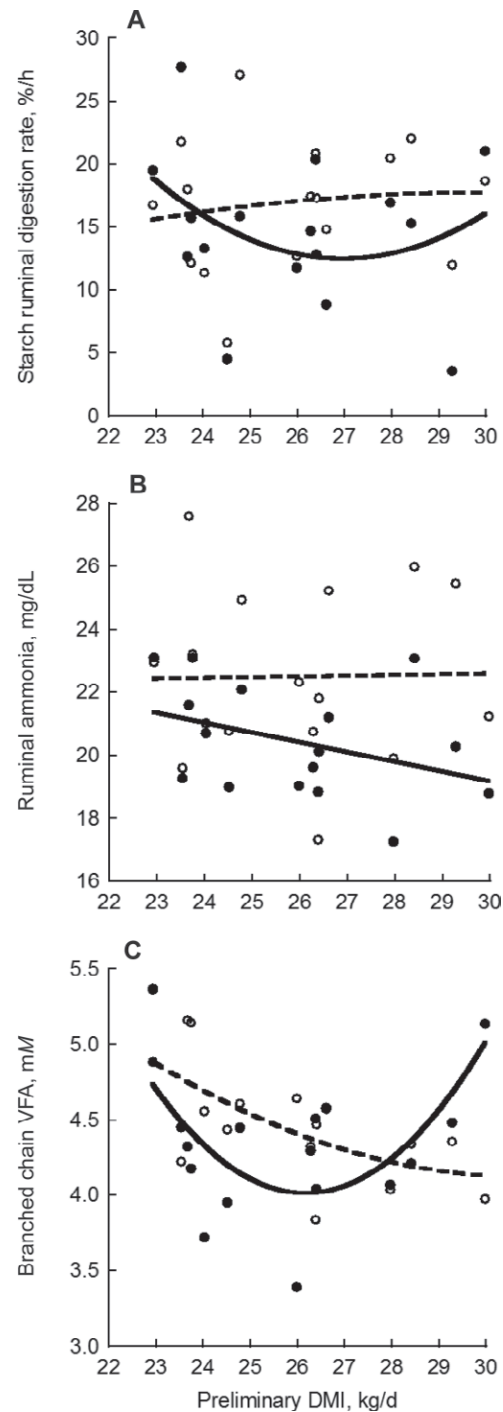


Figure 4. Interaction of early (open circles, dashed line) and late (closed circles, solid line) alfalfa maturity with preliminary DMI for A) starch digestion rate (interaction: $P = 0.13$ quadratic; early: $P = 0.88$, $R^2 = 0.02$; late: $P = 0.47$, $R^2 = 0.11$), B) ruminal ammonia concentration (interaction: $P = 0.14$ linear; early: $P = 0.95$, $R^2 = 0.001$; late: $P = 0.15$, $R^2 = 0.14$), and C) branched-chain VFA concentration (interaction: $P = 0.01$ quadratic; early: $P = 0.03$, $R^2 = 0.43$; late: $P = 0.04$, $R^2 = 0.39$). The preliminary DMI on the x-axis are the mean DMI of individual cows during the final 4 d of the preliminary period when all cows were fed a common diet. The best-fit lines are drawn to demonstrate the significant interaction even if the individual relationships are not significant.

dent of treatment. This is in agreement with the results in a review by Clark et al. (1992) who reported positive linear relationships between OM intake and fluxes of NAN, NANMN, and MN as OM intake increased over a very wide range (3 to 23 kg/d).

Based on studies with continuous-culture fermenters, increases in solid and liquid dilution rates, which can be associated with increased intake, resulted in greater microbial efficiency (Crawford et al., 1980; Shriver et al., 1986). In this experiment, microbial efficiency tended to be related to pDMI (interaction $P = 0.14$; Table 11), but the response differed by treatment. The greatest difference in microbial efficiency was for cows with high pDMI, such that EC had lower microbial efficiency than LC (Figure 5). Microbial efficiency of cows consuming EC quadratically decreased as true ruminally digested OM increased ($P = 0.03$, $R^2 = 0.42$; Figure 6); however, no relationship existed between microbial efficiency and true ruminally digested OM for cows consuming LC ($P = 0.65$). This is the opposite of that demonstrated in the experiment of similar design previously mentioned that evaluated the effects of grass maturity; no relationship existed between microbial efficiency and true ruminally digested OM for cows consuming diets with

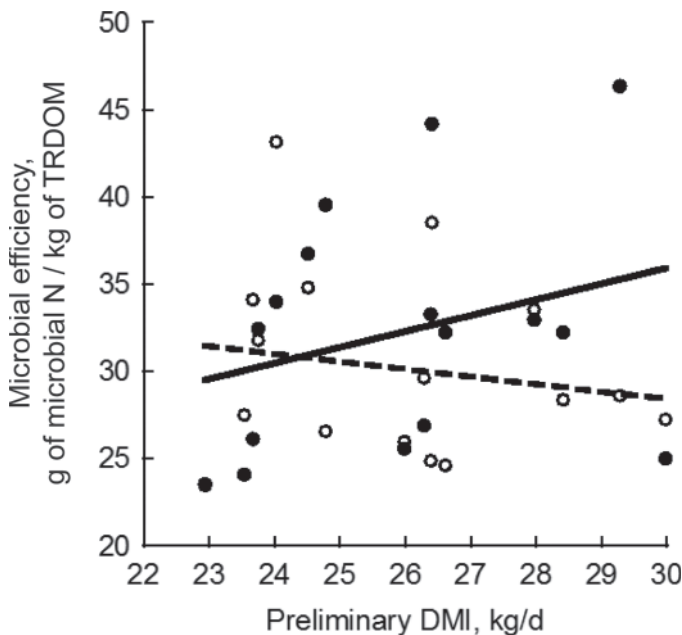


Figure 5. Interaction of early (open circles, dashed line) and late (closed circles, solid line) alfalfa maturity with preliminary DMI for microbial efficiency (interaction: $P = 0.14$ linear; early: $P = 0.52$, $R^2 = 0.03$; late: $P = 0.29$, $R^2 = 0.08$) expressed as g of microbial N/kg of true ruminally digested OM (TRDOM). The preliminary DMI on the x-axis are the mean DMI of individual cows during the final 4 d of the preliminary period when all cows were fed a common diet. The best-fit lines are drawn to demonstrate the significant interaction even if the individual relationships are not significant.

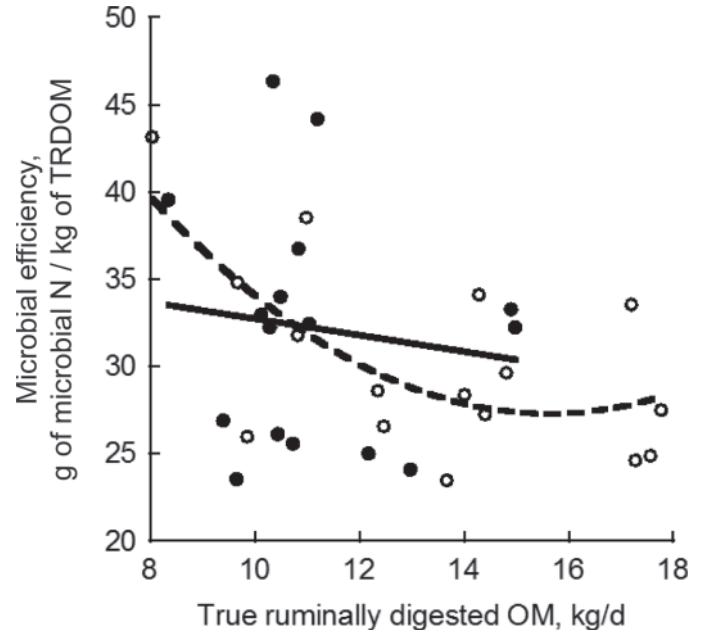


Figure 6. Relationship between true ruminally digested OM (TRDOM) and microbial efficiency for early {open circles, dashed line; microbial efficiency, g of microbial N/kg of TRDOM = $47.3 - (1.43 \times \text{TRDOM, kg/d}) + [0.213 \times (\text{TRDOM, kg/d} - 12.3)^2]$; $P = 0.03$, $R^2 = 0.42$ } and late (closed circles, solid line; $P = 0.65$, $R^2 = 0.02$) alfalfa maturity.

early-cut grass silage and a linear decrease in microbial efficiency as true ruminally digested OM increased for cows consuming diets with late-cut grass silage (K. L. Kammes and M. S. Allen, unpublished data). Additionally, microbial efficiency decreased linearly, independent of treatment, as ruminal digestion rate of starch increased ($P = 0.002$; Figure 7). Similarly, others have reported that efficiency of MN production was inversely related to true ruminally digested OM (kg/d; Oba and Allen, 2003b) and starch digestion rate (Oba and Allen, 2003b; Voelker and Allen, 2003) in experiments evaluating carbohydrate source, concentration, and fermentability of diets in dairy cattle. These negative relationships indicated that factors other than availability of energy limited efficiency of MN production and energy from OM fermentation was uncoupled from microbial growth (Russell and Cook, 1995).

CONCLUSIONS

Delaying the harvest of AL resulted in lower CP and higher NDF and iNDF concentrations. Alfalfa maturity and its interaction with pDMI did not affect milk or FCM yields but EC increased DMI compared with LC; thus, EC had lower efficiency of milk production than LC. The EC diet decreased milk fat concentration more per kilogram of pDMI increase than did the LC

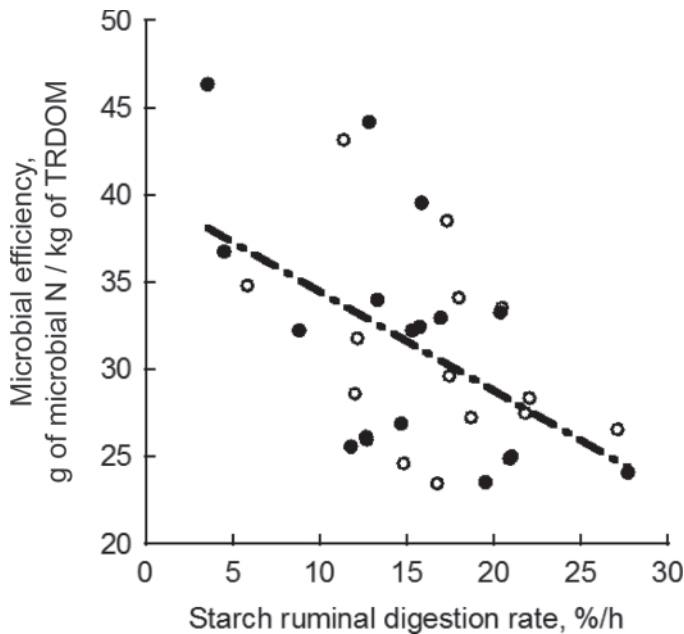


Figure 7. Relationship between starch ruminal digestion rate and microbial efficiency. Microbial efficiency, g of microbial N/kg of true ruminally digested OM (TRDOM) = $40.1 - 0.568 \times$ starch ruminal digestion rate, %/h ($P = 0.002$, $R^2 = 0.27$). Open circles denote early and closed circles denote late alfalfa maturity.

diet, but milk fat yield was not affected. The lower concentration and faster passage rate of iNDF for EC resulted in lower rumen pools of iNDF, total NDF, and DM for EC compared with LC, which EC increased at a slower rate than did LC as pDMI increased. The EC diet decreased starch intake and increased ruminal pH compared with the LC diet. Rate of ruminal starch digestion was related to level of intake; however, this did not affect ruminal or postruminal starch digestion. Total-tract digestibility of NDF, OM, and DM was higher for EC than for LC. When AL silage was the only source of forage in the diet, cows supplemented with additional concentrate to account for decreased protein and increased fiber concentrations associated with LC produced similar FCM yields with greater efficiency than cows fed EC.

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